

Defence Preparation Guide

What changed since the 21 September draft, why, and how to defend it

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How to use this guide

Part 1 lists every substantive change made since the draft Prof. Oyewande reviewed on 21 September, with the old and new statement side by side. Part 2 explains the physics and statistics behind the changes at the depth a panel is likely to probe. Part 3 is a bank of likely questions with answers you can defend. Part 4 collects the numbers you should know by heart and tells you where each result lives in the repository.

Read Part 2 slowly, with the notebooks open. Every claim in it was checked either against the primary papers or by running the code; the places where something was *not* re-verified are flagged explicitly in Section 2.9.

1 What changed since 21 September

1.1 Theory

21 September draft	Current version
Three coefficients derived: b_μ , $H_{\mu\nu}$, $d_{\mu\nu}$.	Four: b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$, $H_{\mu\nu}$, shown to be the complete set producing spin-dependent terms.
d_{ij} and d_{00} “correct in structure but carry an open overall sign”; objective counted as four of five achieved.	Sign resolved: Kostelecký and Lane’s lower-index p_j is the covariant component $p_j = -p^j$. All three $d_{\mu\nu}$ components match exactly. All five objectives met.
Limitation: “only three of the eight SME coefficients derived”.	Removed. a_μ , $c_{\mu\nu}$, e_μ enter only spin-independent terms; f_μ does not enter at this order and can be removed by a field redefinition (Altschul 2006).
e_μ , f_μ , $g_{\lambda\mu\nu}$ called “dimension-5”.	They are dimensionless. The correct point (Colladay and Kostelecký 1998) is that no renormalisable, gauge-invariant term of the full SME generates them, so they are expected to be suppressed, though possibly not for nucleons (Kostelecký and Lane 1999a).
Antiparticle Hamiltonian taken as minus the lower block at the same $\mathbf{p}$ and $\boldsymbol{\sigma}$.	Antiparticle compared as the CPT-conjugate state, with the momentum reversed: $H^{\bar{f}}(\mathbf{p}, \boldsymbol{\sigma}) = -H_{\text{lower}}(-\mathbf{p}, \boldsymbol{\sigma})$. Verified for every SME coefficient family.
Master table last column: “ $ A_\alpha \rightarrow 1$ predicted” for CPT-odd rows, “ $A_\alpha \approx 0$ ” for CPT-even rows.	Replaced by the actual relation for CPT-conjugate states, $g^{\bar{f}} = -g^f$ (CPT-odd) or $g^{\bar{f}} = g^f$ (CPT-even). The old entries contradicted the thesis’s own divergence result.
Both Kostelecký–Lane papers cited as “1999”.	Split into 1999a (PRD 60, 116010) and 1999b (J. Math. Phys. 40, 6245), each cited where used. Altschul (2006) and Wu <i>et al.</i> (2023) added.

1.2 Database and results

21 September draft	Current version
273 datasets; 21 classified UNKNOWN.	283 compiled, 247 analysed. The 36 held out are itemised: 28 combined envelopes (12 $g_A g_V$, 16 $g_p g_s$), 5 duplicates at 90% CL, 2 template placeholders, 1 bound with no potential assignment.
19 of 273 datasets involve antimatter.	22 of 247. Muonium ($e^- \mu^+$) bounds moved to the antimatter side, consistent with how $dd\mu^+$ was already treated.
Ten matched pairs.	Fifteen, of which thirteen are independent (two $g_V g_V / g_p g_p$ rows use byte-identical Fadeev files).
Six of eleven potentials, four of nine sectors with no antimatter data.	Seven of twelve potentials; six of eleven sector pairs.
Priority sectors $e-N$, $e-n$, $n-p$, $n-n$.	$e-N$, $n-N$, $p-N$, $e-n$, $n-p$, $n-n$ (150 of the 225 matter-sector datasets).
$g_s g_s$ pair described as a “CPT-even channel”.	Described as spin-independent; a_μ (CPT-odd), $c_{\mu\nu}$ and e_μ (CPT-odd) all feed spin-independent terms, so “CPT-even” could not be defended.

1.3 Statistics

21 September draft	Current version
“Every pair remains significant at effectively $p \rightarrow 0$.”	Bug fixed: the full χ^2 had been compared with the reduced degrees of freedom. Now $\chi_{\text{eff}}^2 = \chi^2 n_{\text{eff}} / n$ on n_{eff} dof, quoted as a Gaussian Z . Range $Z = 5.1$ to 51.
Bootstrap interval on mean $ A_\alpha $ by delta-method, clipped at 1.	Couplings perturbed log-normally, A_α recomputed exactly, median bootstrap bias removed. Every interval now contains its point estimate (before, six did not).

1.4 Errors found in the final review and corrected

- The printed $H_{\text{NR}}(g_{[0k]0})$ and $H_{\text{NR}}(g_{[kl]j})$ had the wrong sign: they were Kostelecký and Lane’s expressions without the $p_j = -p^j$ conversion. Correct forms: $+\boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{g}_{00})$ and $+\frac{1}{2}\varepsilon_{klm}g_{mlj}p^j\sigma^k$.
- “0.9998, the highest asymmetry”: the highest is 1.0000.
- “Every Cong 2025 pair sits at the top of the table”: false; replaced by a like-for-like comparison (all four rose).
- $e-N$ given as 54 in the text but 50 in the table; “273 rows” should be 283; “16–31” datasets should be 14–31; atlas caption said 25 curves from 10 experiments, the figure shows 27 from 11.
- Unit-error pairs: three matched pairs touch an affected dataset, not two.
- Delaunay *et al.* (2017) was listed as an antimatter experiment; it is hydrogen and helium spectroscopy.

- The $\bar{p}$ -He $|\alpha|$ bound was attributed to Ficek *et al.* (2018); it comes from Salumbides *et al.* (2014).
- The tensor current $\bar{\psi}\sigma^{\mu\nu}\psi$ was called C-even in the notes; it is C-odd (checked numerically).

2 The physics and statistics, in depth

2.1 The minimal SME fermion Lagrangian

For one Dirac fermion (Kostecký and Lane 1999a, Eqs. 1–3),

$$\mathcal{L} = \frac{1}{2}i\bar{\psi}\Gamma^\nu\overleftrightarrow{\partial}_\nu\psi - \bar{\psi}M\psi, \quad (1)$$

$$M = m + a_\mu\gamma^\mu + b_\mu\gamma_5\gamma^\mu + \frac{1}{2}H_{\mu\nu}\sigma^{\mu\nu}, \quad (2)$$

$$\Gamma^\nu = \gamma^\nu + c^{\mu\nu}\gamma_\mu + d^{\mu\nu}\gamma_5\gamma_\mu + e^\nu + if^\nu\gamma_5 + \frac{1}{2}g^{\lambda\mu\nu}\sigma_{\lambda\mu}. \quad (3)$$

Index counting. Each coefficient carries the indices of its Dirac matrix plus, in Γ^ν , the derivative index ν . Scalar and pseudoscalar structures carry none, vector and axial vector one, the tensor σ two. So e, f have one index, c, d two, and g three, which is why it needs a third letter, λ . It is antisymmetric in its first two indices because $\sigma^{\lambda\mu}$ is.

Dimensions. “The parameters appearing in M have dimensions of mass, while those in Γ are dimensionless” (Kostecký and Lane 1999a). So a, b, H have mass dimension 1 and c, d, e, f, g are dimensionless.

Why e, f, g are special. Colladay and Kostecký (1998, near their Eq. 29): all renormalisable terms that could generate them “are directly incompatible with the electroweak structure”; they could come only from higher-dimensional operators involving the Higgs field, so they are expected to be highly suppressed. Kostecký and Lane (1999a) add that for protons and neutrons, strong binding “might generate effective terms governed by appreciable parameters”. That caveat is the physical reason studying $g_{\lambda\mu\nu}$ is worthwhile for nucleon sectors.

CPT parity from bilinears. The CPT parities of $\bar{\psi}\Gamma\psi$ are + (scalar), + (pseudoscalar), – (vector), – (axial vector), + (tensor); each derivative adds a –. That gives a, b odd; H even; c, d even; e, f, g odd, exactly Kostecký and Lane’s classification.

2.2 The Foldy–Wouthuysen reduction

Split the Dirac Hamiltonian into even (block-diagonal) and odd (off-diagonal) parts relative to β . The FW generator $S = -i\beta\mathcal{O}/2m$ removes the odd part order by order:

$$H' \approx \beta m + \mathcal{E} + \frac{\beta\mathcal{O}^2}{2m} - \frac{[\mathcal{O}, [\mathcal{O}, \mathcal{E}]]}{8m^2} + \dots \quad (4)$$

An even perturbation survives at order m^0 (e.g. b_i, H_{ij}). An odd perturbation contributes only through its cross term with $\boldsymbol{\alpha} \cdot \mathbf{p}$, at order $1/m$ (e.g. b_0, H_{0i}). Kinetic coefficients multiply p_ν , and when $\nu = 0$ that brings in the energy, $\approx m$, which is why d_{i0} and $g_{[kl]0}$ are mass-enhanced (order m^{+1}).

Coefficients in Γ_0 first need the field redefinition $A = 1 - \frac{1}{2}\gamma^0(\Gamma_0 - \gamma_0)$ (Kostecký and Lane 1999b) so that the time-derivative term is standard and the Hamiltonian is Hermitian.

2.3 The momentum-index convention (the old “open sign”)

Kostecký and Lane’s non-relativistic Hamiltonian (1999a, Eq. 4) is written with lower-index p_j . Their intermediate relativistic Hamiltonian (1999b) has the free term $m\mathcal{P}_0 := -p_j\gamma^0\gamma^j$. That

equals $-\boldsymbol{\alpha} \cdot \mathbf{p}$ if p_j is read as the physical momentum, which would give the wrong free Dirac Hamiltonian. To recover $+\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m$, p_j must be the covariant component, $p_j = -p^j$. Under that reading every $d_{\mu\nu}$ term agrees, signs included. The mass-enhanced d_{i0} term has no momentum, which is why it agreed all along while the momentum-linear terms seemed flipped: that pattern is the fingerprint of an index convention, not of an error. The $g_{\lambda\mu\nu}$ sector, derived independently, agrees under the same reading, an independent check.

2.4 Completeness of the four coefficients

In Kostecký and Lane’s Hamiltonian to third order (1999b, Eq. 26), every term multiplied by $\boldsymbol{\sigma}$ contains only b, d, g, H . The spin-independent terms contain a, c, e ; f appears nowhere. Since every Dobrescu–Mocioiu potential except V_1 is spin-dependent, the four coefficients are the complete spin-dependent dictionary at these orders. Altschul (2006) shows f_μ can be removed entirely by a field redefinition.

2.5 Matter versus antimatter: the central subtlety

After FW reduction the even Hamiltonian has an upper block (particles) and a lower block (negative-energy solutions).

Hole theory. An antiparticle of momentum $\mathbf{p}$ and spin $\mathbf{s}$ is the *absence* of a negative-energy solution with momentum $-\mathbf{p}$ and spin $-\mathbf{s}$, and its energy is minus that solution’s energy.

Two legitimate comparisons.

1. *CPT-conjugate state.* CPT maps a particle $(\mathbf{p}, \mathbf{s})$ to an antiparticle $(\mathbf{p}, -\mathbf{s})$ (C swaps particle and antiparticle, P reverses $\mathbf{p}$, T reverses $\mathbf{p}$ and $\mathbf{s}$). Labelling that state by the particle’s spin operator:

$$H^{\bar{f}}(\mathbf{p}, \boldsymbol{\sigma}) = -H_{\text{lower}}(-\mathbf{p}, \boldsymbol{\sigma}). \quad (5)$$

Under exact CPT symmetry this equals the particle Hamiltonian, so this is the comparison under which $A_\alpha = 0$. The thesis defines $g^{\bar{f}}$ this way.

2. *Same physical spin.* Writing $H_{\text{lower}} = a_0 + \mathbf{a} \cdot \boldsymbol{\sigma}$, charge conjugation ($\chi \rightarrow i\sigma_2\chi^*$) gives

$$H_{\text{same spin}}^{\bar{f}}(\mathbf{p}, \boldsymbol{\sigma}) = -a_0(-\mathbf{p}) + \mathbf{a}(-\mathbf{p}) \cdot \boldsymbol{\sigma}. \quad (6)$$

The rule. Between CPT-conjugate states a coefficient flips sign exactly when it is CPT-odd. At the same physical spin, the relation is (CPT parity) $\times$ (parity of the term under $\mathbf{s} \rightarrow -\mathbf{s}$), counting even as $+1$ and odd as -1 , where $+1$ means the same sign and -1 a flip. For b : $(-1)(-1) = +1$, same sign; for H : $(+1)(-1) = -1$, a flip. The two differ only for spin-linear terms, and that difference is Kostecký and Lane’s antiparticle rule (1999b, last paragraph): $a \rightarrow -a$, $b \rightarrow +b$, $c \rightarrow +c$, $d \rightarrow -d$, $e \rightarrow -e$, $f \rightarrow -f$, $g \rightarrow +g$, $H \rightarrow -H$.

Worked example: b_i . $\gamma^0\gamma_5\gamma^i = -\Sigma^i$ and $\Sigma^i = \text{diag}(\sigma^i, \sigma^i)$, so both blocks equal $-\mathbf{b} \cdot \boldsymbol{\sigma}$. With no momentum dependence, $H_{\text{CPT}}^{\bar{f}} = -(-\mathbf{b} \cdot \boldsymbol{\sigma}) = +\mathbf{b} \cdot \boldsymbol{\sigma}$: opposite to the particle. At the same physical spin the antiparticle has $-\mathbf{b} \cdot \mathbf{s}$, the same as the particle. Both statements are true; they compare different states.

Why C alone does not do it. The axial current is C-even (checked: $C(\gamma_5\gamma^\mu)^T C^{-1} = +\gamma_5\gamma^\mu$), so charge conjugation of the bilinear gives no sign. The flip is a CPT effect.

What went wrong before. The earlier method took $-H_{\text{lower}}(\mathbf{p}, \boldsymbol{\sigma})$, reversing neither momentum nor spin. It agrees with the CPT-conjugate comparison only for terms even in $\mathbf{p}$. For b_i that is harmless, which is why the central result survived. For $g_{[0k]0}$ and $g_{[kl]j}$, which are linear in $\mathbf{p}$, it produced the false claim that only one g family flips.

External checks. Bluhm, Kostelecký and Russell (1999): their hydrogen-to-antihydrogen substitution leaves b_μ unchanged (the same-spin rule), and the b -dependent 1S–2S shift reverses because “the hyperfine states in $\bar{\text{H}}$ have opposite positron and antiproton spin assignments”: the compared states are CPT conjugates. `FW_antiparticle_all_coefficients.ipynb` checks both comparisons for all 16 coefficient families; both agree with CPT parity and with the published rule in every case.

2.6 The asymmetry parameter and why it cannot test CPT with bounds

$A_\alpha = (g^f - g^{\bar{f}})/(g^f + g^{\bar{f}})$, with $g^{\bar{f}}$ for the CPT-conjugate state.

- Exact CPT symmetry: $g^{\bar{f}} = g^f$, so $A_\alpha = 0$.
- A CPT-odd coefficient alone: $g^{\bar{f}} = -g^f$, so the denominator vanishes and A_α diverges. It does *not* approach 1.
- Published bounds are positive numbers $|g| \leq B$. Put two of them in: $A_\alpha = (1 - r)/(1 + r)$ with $r = B_{\bar{f}}/B_f$. It depends only on the ratio of sensitivities. One bound much tighter than the other gives $|A_\alpha| \rightarrow 1$ whatever the physics.
- Any true pair of couplings below the two bounds is allowed, so the true A_α can be anything. The bounds do not constrain it.

Empirical confirmation. The four pairs built on the 2025 hydrogen bounds of Cong *et al.* each have an older twin with the same coupling, potential and antimatter partner. Tightening only the matter side raised the asymmetry every time: $0.9998 \rightarrow 0.9999$, $0.8044 \rightarrow 1.0000$, $0.8236 \rightarrow 0.9303$, $0.7994 \rightarrow 0.9304$.

Design criterion. A new antimatter bound is useful for a CPT test in proportion to how well its sensitivity matches the matter bound, not how tight it is. That is why the Jiao/Karshenboim pair (0.3336), the least saturated, is the most informative.

2.7 Statistics

Grid. Both curves are interpolated log-linearly onto 300 log-spaced points over their overlap.

Uncertainties. Published exclusion curves carry no error bars. Each point gets a fractional uncertainty from the local curvature of the curve in log-log space, between 2% and 50%. This is a heuristic, and the χ^2 values depend on it; the conclusions do not.

Weighted χ^2 . $\chi^2 = \sum_i (g_{m,i} - g_{a,i})^2 / (\sigma_{m,i}^2 + \sigma_{a,i}^2)$ tests H_0 : the two exclusion curves coincide on the grid. Rejecting it means the two experiments reached different sensitivities. It says nothing about the couplings themselves.

Autocorrelation. Adjacent grid points are not independent. The autocorrelation length τ is the lag where the normalised autocorrelation of the χ^2 residuals first drops below $1/e$; $n_{\text{eff}} = n/\tau$. The statistic is rescaled by the same factor, $\chi_{\text{eff}}^2 = \chi^2 n_{\text{eff}}/n$, and compared with a χ^2 distribution on n_{eff} dof. (The old code compared the full χ^2 with n_{eff} , which overstated significance.) Tail probabilities are computed in arbitrary precision and quoted as a one-sided Gaussian Z . Example: Jiao/Karshenboim, $\chi_{\text{eff}}^2 = 48.5$ on 9 dof, $p \approx 2 \times 10^{-7}$, $Z = 5.1$.

Bootstrap. 2000 resamples; each perturbs both couplings by $g \rightarrow g e^{\sigma\epsilon}$ and recomputes A_α exactly. $|A_\alpha|$ is folded at 0 and saturates at 1, so resampled means sit off the estimate; the median offset (the bootstrap bias) is subtracted before quoting the 95% interval. An interval excluding zero is not, on its own, evidence of asymmetry, because $|A_\alpha|$ is biased away from zero by construction.

2.8 The database

- Sources: the compilation of Cong *et al.* (RMP 2025) and the primary papers. Filename prefixes give the potential: 3 means V_3 alone, 23 means $V_2 + V_3$.
- Units normalised to metres (m, MeV^{-1} , eV^{-1} , cm, nm, fm). $|\alpha|$ (gravity-normalised) and $|g|$ bounds are kept apart; typical values differ by more than thirty orders of magnitude.
- Muonium: all six $e\text{-}\mu$ bounds come from muonium (Karshenboim’s μe curve is from muonium hyperfine data; Ohayon 2022 and Stadnik 2023 are $e^-\mu^+$). Muonium contains an antimuon, so these are antimatter.
- Six datasets have implausible ranges up to 10^{52} m (two V_3 , two V_{9+10} , two V_{15}). Three matched pairs touch one, but the tails lie outside every overlap used.
- 48 datasets are the same bound filed under several couplings; the coverage figures collapse them to 199 independent bounds.

2.9 What was not re-verified

- “Every potential assignment agrees with the source compilation’s records for the 194 datasets they cover.” This comes from an earlier check against the Cong *et al.* online records (Zenodo 10.5281/zenodo.14572652) and was not repeated.
- Why the source compilation files the Fadeev *et al.* V_3 curves under both $g_V g_V$ and $g_p g_p$. The Fadeev paper treats pseudovector and pseudoscalar interactions; the $g_V g_V$ label comes from the compilation. The thesis only states the fact that the rows are identical.

3 Likely questions and answers

3.1 Scope and contribution

Q1. In one sentence, what does this thesis do?

Answer. It builds a complete dictionary from the spin-dependent SME coefficients to the Dobrescu–Mocioiu potentials, compiles the published bounds into one standardised database, and shows what a matter–antimatter comparison of those bounds can and cannot say about CPT symmetry.

Q2. What is new here? Kostelecký and Lane already derived the non-relativistic Hamiltonian.

Answer. Three things. The explicit matching of each SME term to the Dobrescu–Mocioiu basis, together with a proof that b, d, g, H are the complete spin-dependent set. A standardised, auditable database of 283 bounds with a matching pipeline. And the result that an asymmetry built from one-sided bounds tracks sensitivity, not CPT status, shown both analytically and in the data.

Q3. Your main result is a limitation. Why is that a contribution?

Answer. Because it is the thing a reader would otherwise get wrong. All fifteen pairs are highly significant; without the analysis one would be tempted to read that as a CPT signal. The thesis shows why it is not, turns that into a concrete design criterion (sensitivity matching), and identifies exactly which sectors and potentials need antimatter data.

Q4. Why these four coefficients and not the others?

Answer. In Kostelecký and Lane’s third-order Hamiltonian, only b , d , g , H appear in spin-dependent terms. a , c , e appear only in spin-independent terms, and f not at all. All potentials except V_1 are spin-dependent, so nothing is missing.

3.2 The SME–DM mapping

Q5. SME coefficients are fixed background fields; DM potentials come from boson exchange between two fermions. How can one map to the other?

Answer. The dictionary matches operator structure, not mechanism. A background b_i gives a single-particle term $-\mathbf{b} \cdot \boldsymbol{\sigma}$; a boson exchange gives two-body terms with the same spin structure at each vertex. The mapping tells you which Dobrescu–Mocioiu structure an experiment must be sensitive to in order to constrain a given SME coefficient combination. It does not claim the two are the same physics. This is a fair question and the honest answer is exactly that.

Q6. Why the Foldy–Wouthuysen transformation?

Answer. It gives the non-relativistic Hamiltonian as an explicit 2×2 Pauli operator, order by order in $1/m$, which is exactly the form needed to compare term by term with the potentials. It is also the method Kostelecký and Lane used, which made a term-by-term check possible.

Q7. What was the d_{ij} sign problem, and how do you know it is solved and not hidden?

Answer. The momentum-linear terms appeared to have the opposite sign to Kostelecký and Lane. Their own free-particle term, $m\mathcal{P}_0 = -p_j \gamma^0 \gamma^j$, only reproduces the correct free Dirac Hamiltonian if $p_j = -p^j$. With that, all terms agree. Two checks make it convincing: the mass term, which has no momentum, always agreed; and the g sector, derived separately, agrees under the same convention. A wavefunction-renormalisation explanation was tested and ruled out.

Q8. Why does g have three indices? Why λ ?

Answer. It multiplies $\sigma_{\lambda\mu}$ (two indices) inside Γ^ν (one more, the derivative index). λ is just the next free letter; it is the SME convention. Note that the thesis also uses λ for the interaction range; context distinguishes them.

Q9. Is $g_{\lambda\mu\nu}$ really dimension five?

Answer. No; that was an error in the earlier draft. It is dimensionless and multiplies a dimension-four operator. What is true is that it cannot come from a renormalisable, gauge-invariant term of the full SME, so it is expected to be suppressed for electrons (Colladay and Kostelecký 1998), but possibly not for nucleons (Kostelecký and Lane 1999a).

Q10. Which g families contribute, and why does $g_{[0k]j}$ vanish?

Answer. $g_{[kl]0}$ (Zeeman-like, mass-enhanced, V_2), $g_{[0k]0}$ (spin-velocity, V_7), $g_{[kl]j}$ (velocity-dependent, V_7, V_8). $g_{[0k]j}$ gives nothing at the orders kept; in Eq. 26 of Kostelecký and Lane it first appears at order p^2/m^2 .

3.3 CPT and antimatter

Q11. Does b_μ flip sign between matter and antimatter or not? Kostelecký and Lane say $b \rightarrow +b$.

Answer. Both are right; they compare different states. Between CPT-conjugate states (antiparticle spin reversed) the b term flips, which is the CPT-odd signature. At the same physical spin it does

not, which is the Kostelecký–Lane rule. They differ because $\mathbf{b} \cdot \boldsymbol{\sigma}$ changes sign when the spin is reversed. The thesis uses the CPT-conjugate comparison because that is the one under which exact CPT symmetry gives $A_\alpha = 0$.

Q12. Why not just use charge conjugation?

Answer. The axial current is C-even, so C alone gives no sign difference for b . The difference is a CPT effect. Charge conjugation of the two-spinor is used, but together with the hole-theory reversal of momentum and spin.

Q13. Your earlier draft said only one g family flips. What happened?

Answer. The antiparticle was taken as minus the lower block without reversing the momentum. That is harmless for terms with no momentum, such as b_i , but wrong for the two momentum-linear g families. With the reversal all three flip, as a CPT-odd coefficient must. I then checked every SME coefficient family both ways in one notebook.

Q14. How do you know your antiparticle rule is right?

Answer. Three independent checks agree: the full symbolic FW reduction with charge conjugation for all 16 families; Kostelecký and Lane’s published antiparticle substitution rule; and Bluhm, Kostelecký and Russell’s antihydrogen analysis. A direct numerical diagonalisation of the Dirac Hamiltonian also gives the same signs.

Q15. Muonium: why is it antimatter?

Answer. Muonium is $e^-\mu^+$; the μ^+ is an antimuon. The database already counted $dd\mu^+$ as antimatter for exactly that reason, so treating muonium as matter was inconsistent.

3.4 The asymmetry parameter

Q16. If A_α built from bounds cannot test CPT, why define it at all?

Answer. It is the natural observable, and with signed measurements of the same coupling in both sectors it would work. With bounds, it becomes a diagnostic of sensitivity matching. Showing precisely where the obvious observable breaks down is part of the result.

Q17. What does a CPT-odd coefficient predict for A_α ?

Answer. For signed couplings, $g^{\bar{f}} = -g^f$ and A_α diverges. For magnitudes it gives $|g^{\bar{f}}| = |g^f|$. Neither gives $|A_\alpha| \rightarrow 1$. Values near 1 in the data come from sensitivity gaps.

Q18. Then why is the lowest-asymmetry pair the most informative?

Answer. Because its two bounds are closest in sensitivity, so it is the only comparison not dominated by the sensitivity ratio.

Q19. The $g_s g_s$ pair: is it CPT-even?

Answer. I do not claim that. V_1 is spin-independent; a_μ and e_μ (CPT-odd) and $c_{\mu\nu}$ (CPT-even) all enter spin-independent terms. What the pair shows is that a channel none of the four spin-dependent coefficients feeds has an asymmetry comparable to the spin-dependent pairs, again pointing to sensitivity, not physics.

3.5 Statistics

Q20. What is your null hypothesis?

Answer. That the matter and antimatter exclusion curves coincide at every grid point, tested

jointly with a weighted χ^2 . Rejecting it means the experiments reached different sensitivities. It is not a test of the underlying couplings.

Q21. What does a p -value mean for an asymmetry built from upper limits?

Answer. Only this: if the two curves were the same curve plus noise of the assumed size, how unlikely is the observed disagreement. It measures a sensitivity difference.

Q22. Where do your uncertainties come from? Published bounds have none.

Answer. From a curvature heuristic, 2% to 50% of the bound. That is a modelling choice. The Z values depend on it; the qualitative conclusions do not, because they rest on the analytic argument and on the like-for-like comparisons, not on the χ^2 .

Q23. Why correct for autocorrelation, and how?

Answer. 300 interpolated points come from far fewer real measurements, so they are correlated. I estimate τ from the residual autocorrelation, set $n_{\text{eff}} = n/\tau$, and rescale χ^2 by the same factor. An earlier version rescaled only the degrees of freedom, which overstated significance; that is fixed.

Q24. Your earlier draft said $p \rightarrow 0$. Now?

Answer. Z from 5.1 (Jiao/Karshenboim, $p \approx 2 \times 10^{-7}$) to 51. The earlier “ $p \rightarrow 0$ ” was partly the bug and partly floating-point underflow.

Q25. Your bootstrap interval excludes zero. Is that evidence of asymmetry?

Answer. No. $|A_\alpha|$ is biased away from zero by the absolute value, so its interval excludes zero almost always. The interval only describes spread from the assumed uncertainties. I also correct the median bootstrap bias, which otherwise pushes intervals off the estimate near 0 and 1.

Q26. Fifteen pairs, but are they independent?

Answer. Thirteen are. Two $g_V g_V / g_p g_p$ rows use byte-identical files, and several pairs share one side (Ficek 2018 is the antimatter side of seven). The shared datasets mean the pairs are not fully independent even among the thirteen, which is another reason not to combine them into a single significance.

3.6 Data

Q27. How were potentials assigned?

Answer. From filename prefixes (the numeric prefix is the potential number), explicit tokens, and a small override table, checked against the source compilation’s records. *Caveat:* the claim of agreement for the 194 datasets those records cover comes from an earlier check I did not repeat before this defence.

Q28. Why exclude 36 datasets?

Answer. 28 are combined envelopes over several experiments with no single potential; 5 are 90% CL duplicates of 95% CL curves; 2 are template placeholders; 1 has no potential that can be established. Each is recorded with its reason; none is silently dropped.

Q29. What about the implausible interaction ranges?

Answer. Six datasets reach up to 10^{52} m, almost certainly a mis-detected unit. Three matched pairs touch one, but the asymmetry is computed only on the overlap, which in all three cases lies far inside the plausible range.

Q30. Why is the $\bar{p}$ - N bound not used?

Answer. It is published as a gravity-normalised strength $|\alpha|$, not a coupling $|g|$; comparing it with the p - N data would need a conversion the framework does not perform.

3.7 Limitations and future work

Q31. What are the main limitations?

Answer. The asymmetry cannot distinguish CPT violation from sensitivity gaps with bounds; the uncertainty model is heuristic; derivations stop at the orders kept in $1/m$; 36 datasets are held out; antimatter coverage is thin, which is the finding itself.

Q32. What would turn this into a real CPT test?

Answer. Antimatter-sector bounds of sensitivity comparable to their matter counterparts, or signed measurements. The highest-value sectors are e - N , n - N and p - N , and the potentials V_{1a} , V_{9+10} , V_{12+13} . Reanalysing existing BASE, ALPHA and co-magnetometer data, as Wu *et al.* (2023) did for Lorentz-violation data, is the cheapest first step.

Q33. What would you do next with more time?

Answer. Carry the reductions to higher order in $1/m$; decompose the 28 envelope curves; re-verify all potential assignments against the source records; add a conversion for $|\alpha|$ bounds; and propagate the curve uncertainties more rigorously than the curvature heuristic.

3.8 Awkward questions

Q34. Did you make mistakes along the way?

Answer. Yes, and they are documented: the open d sign, the antiparticle method, a wrong dimension label, a statistics bug and a mis-classified sector. Each was found by checking against primary sources or by rerunning the code, and each fix is recorded with the check that confirms it.

Q35. Is your software trustworthy?

Answer. It has an automated test suite (140 tests), including regression tests for each bug fixed, and the full analysis reruns from one command. The derivations are in executable SymPy notebooks that assert their own results against Kostelecký and Lane.

Q36. Could a referee reproduce Table II from your repository?

Answer. Yes: run the pipeline, and `results/tables/asymmetry_summary.csv` contains every number in the table, including n_{eff} , Z and the intervals.

4 Quick reference

4.1 Numbers to know

Datasets compiled / analysed / held out	283 / 247 / 36
Antimatter / matter datasets (analysed)	22 / 225
Antimatter by sector	$e\bar{p}$ 9, $e\bar{\mu}$ 6, ee^+ 5, $\bar{p}\text{He}$ 1, $dd\mu^+$ 1
Matched pairs (independent)	15 (13)
Pairs in V_2 or V_3	14 of 15
Mean $ A_\alpha $ range	0.3336 to 1.0000
n_{eff} range	6 to 21 (of 300)
Z range	5.1 to 51
Potentials with no antimatter data	7 of 12
Sector pairs with no antimatter data	6 of 11
e - N / n - N / p - N matter datasets	50 / 49 / 18 (117 of 225)
Independent bounds after de-duplication	199
Implausible-range datasets	6

4.2 The fifteen pairs

Coupling	V	Sector	Matter	Antimatter	$ A_\alpha $	n_{eff}	Z
g_{AGA}	3	$e\bar{p}$	Cong 2025	Fadeev 2022	1.0000	8	51.1
g_{AGA}	2	$e\bar{p}$	Cong 2025	Ficek 2018	0.9999	21	34.4
g_{AGA}	2	$e\bar{p}$	Karshenboim 2011	Ficek 2018	0.9998	17	38.8
g_{AGA}	2	ee^+	Ficek 2017	Karshenboim 2011	0.9892	21	48.2
g_{AGA}	3	ee^+	Ficek 2017	Fadeev 2022	0.9539	6	38.5
$g_V g_V$	3	ee^+	Fadeev 2022	Fadeev 2022	0.9535	6	37.1
$g_p g_p$	3	ee^+	Fadeev 2022	Fadeev 2022	0.9535	6	37.1
$g_p g_p$	3	$e\bar{p}$	Cong 2025	Ficek 2018	0.9304	6	34.3
g_{AGA}	3	$e\bar{p}$	Cong 2025	Ficek 2018	0.9303	6	34.3
$g_s g_s$	1	ee^+	Delaunay 2017	Adkins 2022	0.8727	21	25.7
g_{AGA}	3	$e\bar{p}$	Fadeev 2022	Ficek 2018	0.8236	6	37.3
g_{AGA}	3	$e\bar{p}$	Fadeev 2022	Fadeev 2022	0.8044	8	43.8
$g_V g_V$	3	$e\bar{p}$	Fadeev 2022	Ficek 2018	0.7994	7	31.8
$g_p g_p$	3	$e\bar{p}$	Fadeev 2022	Ficek 2018	0.7994	7	31.8
g_{AGA}	2	ee^+	Jiao 2020	Karshenboim 2011	0.3336	9	5.1

4.3 Matter–antimatter relations

Coefficient	CPT	CPT-conjugate state	Same physical spin
b_μ	odd	flips	same
$g_{\lambda\mu\nu}$ (all 3 families)	odd	flips	same
$H_{\mu\nu}$	even	same	flips
$d_{\mu\nu}$	even	same	flips
a_μ, e_μ (spin-independent)	odd	flips	flips
$c_{\mu\nu}$ (spin-independent)	even	same	same

4.4 Where things live

- Derivations: `derivations/sympy/FW_bmu_term.ipynb`, `FW_Hmunu_term.ipynb`, `FW_dmunu_term.ipynb`, `FW_gmunu_term.ipynb`.

- Antiparticle check for every coefficient: `derivations/sympy/FW_antiparticle_all_coefficients.ipynb`.
- Theory notes: `docs/theory_notes/`.
- Pipeline: `spindep_framework`; results in `results/tables/asymmetry_summary.csv` and `dataset_registry.csv`; statistics in `spindep/src/statistics.py`; tests in `tests/`.
- Thesis: `thesis/project/`; journal draft: `thesis/aps-draft/`; summit abstract: `thesis/aps-draft/summit-abstract.txt`.

4.5 Key references

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